

False Consensus: Why Intermediate Agreement Is Not a Safe Early-Exit Signal for Reasoning LLMs

Anonymous ACL submission

Abstract

A popular way to cut reasoning-model inference cost is to probe the partial trajectory for its current answer and stop once probes *agree*. We show this is unsafe as a stopping criterion on its own, and explain why. Self-consistency earns its guarantee from *independently sampled* reasoning paths; probing one trajectory reads the same prefix repeatedly, so agreement carries no vote that can outweigh an error. Hand-labelling 134 wrong stops shows what agreement actually records: three in five commit to an answer the model had not converged on—a placeholder the probe compelled it to emit—and only one in five to an answer it had genuinely settled on. The cost is directional: a stop destroys a correct-in-the-end answer far more often than it rescues a wrong one (up to $\sim 45\times$ at an aggressive latest-probe stop, $\sim 2\times$ at the largest windows, where the rule has almost stopped firing). No tuning escapes this: on frozen trajectories from two models and three math benchmarks, a *preregistered* sweep of the consensus signal (3,520 rules, acceptance gates fixed in advance) clears **no gate**. The failure is specific to the signal—swept identically, a boundary-confidence method (DEER) **clears all three gates** and holds on a held-out split and two unseen models. Early exit is possible; agreement alone cannot make it safe. We release the protocol, the frozen trajectories, and the full sweep.

1 Introduction

Reasoning language models (DeepSeek-AI, 2025; Qwen Team, 2025; OpenAI, 2024) can spend thousands of tokens on a single answer. *Early exit*—halting a chain of thought once its answer is settled—could cut that cost (Chen et al., 2024; Fu et al., 2024), but only if the stopping signal is safe. A widely used signal is *intermediate answer consensus*: at regular intervals a partial trajectory is probed for its current answer (e.g., by appending

a short “Final Answer:” suffix) and generation is stopped once recent probes *agree* (Fu et al., 2024). The underlying assumption is simple: if the model repeatedly says x , it has probably finished reasoning about x . This paper asks whether that assumption is warranted: **is intermediate agreement a safe signal on which to stop?**

Our answer is **no**, and the reason is not a badly tuned threshold. Intermediate consensus¹ forms *while the answer is still moving*. On a frozen replay, a consensus stop destroys a correct-in-the-end answer up to $\sim 45\times$ more often than it rescues a wrong one at an aggressive latest-probe stop. Demanding more agreement helps only by stopping less often: at the largest windows the ratio falls to $\sim 2\times$, but the rule fires on a fifth as many problems and saves almost nothing.

We make this hard to dismiss as “you did not tune it well” by searching the space exhaustively under commitment: on frozen trajectories from two models and three benchmarks we run a *preregistered* sweep of the consensus signal—a window size $\times$ share threshold family plus operational knobs, 3,520 rules—with problem-grouped splits and acceptance gates fixed in advance (Sections 3–4). The outcome does not depend on any single fragile estimate: **not one of the 3,520 rules is both safe and saving—none clears any gate**, and even large windows only trade the drop away for zero saving.

Why should agreement fail so consistently? Section 5 gives three pieces of evidence. Agreement is informative in self-consistency because the reasoning paths are sampled *independently*; probes of one trajectory share a prefix, so they cannot out-vote each other’s errors. Probing the same frozen prefix with two different wordings returns different

¹“False consensus” here denotes non-terminal intermediate probe agreement, unrelated to the social-psychology term (Ross et al., 1977); we study answer-level, not token/layer-level, early exit (Graves, 2016; Schuster et al., 2022).

answers 57% of the time in the first tenth of a trajectory, against 11% at the end—early “answers” are largely artifacts of being asked. And hand-labelling 134 wrong stops finds that three in five commit to an answer the model had not converged on, against one in five that it had genuinely settled on.

To show this is the *signal* and not early exit, we sweep one non-consensus method—DEER, which reads the model’s confidence at a reasoning boundary—through the identical pipeline and gates: it clears all three, losing as little as 0.3 pp while saving 28%. The consensus frontier reproduces on a held-out test split ($r=0.98$) and on unseen models at $4\times$ the scale and a different architecture (Section 4). Two published-signal checks bracket the contrast: CertaIndex (CoT), reproduced from its released implementation, loses 56–70 pp on the same trajectories, while DEER clears the gates on the unseen models too. Figure 1 is the whole paper in one picture. Our claim is about agreement used *alone* as a stopping criterion; combined with a signal that reflects the model’s own estimate it may still be useful, and we return to this in Section 5.

Contribution. Our contribution is an explanation of *why* intermediate answer-consensus cannot serve as an early-exit signal on its own, together with the evidence that no tuning escapes it. Three components support the claim:

- **An account of what probe agreement can and cannot mean.** Self-consistency’s guarantee comes from independently sampled paths; probes of a single trajectory share a prefix, so agreement supplies no vote that can outweigh an error, and a stop has nothing to protect it (Section 5).
- **Direct evidence of what the probes agree on.** Two annotators labelled all 134 stopped-but-wrong cases ($\kappa=0.82$): three in five commit to an answer the model had not converged on, and only one in five to a settled one. Probing forces an output rather than reading a belief, which also explains why a longer window reduces the damage without removing it—and why it does so only by declining to stop (Section 5).
- **A preregistered search showing this is not a tuning failure** (frozen trajectories, offline re-probing, a two-parameter consensus family enumerated to 3,520 rules, gates fixed in advance): *no* consensus rule is both safe and

token-saving; the frontier reproduces on a held-out split and two unseen models; and a non-consensus signal (DEER) swept under the identical protocol *clears the same gates* on the same trajectories, localizing the failure to the signal (Sections 3–4). We release the protocol, trajectories, and sweep in full.

2 Related Work

Probe-based / consensus early exit. The methods we scrutinize periodically query a partial reasoning trajectory for its current answer and halt on agreement. Dynasor / CertaIndex (Fu et al., 2024) formalize this with a “certainty index” over recent probe answers and use it to terminate early when serving reasoning programs, one of several responses to the *overthinking* of reasoning models (Chen et al., 2024; Yang et al., 2025). We adopt the same probe mechanism (a short boxed-answer suffix) and notion of probe agreement, and ask how far *any* rule in this family can be pushed before it becomes unsafe.

Where agreement gets its meaning. Self-consistency (Wang et al., 2023) samples a *diverse* set of reasoning paths and takes the majority answer, on the premise that a hard problem admits many routes to one correct answer. Agreement is evidence there because the paths are drawn independently: when routes that could have diverged do not, that is information about the answer, and the vote can outweigh an error on any single path. Early-stopping variants keep this structure and only decide how many samples to draw (Aggarwal et al., 2023; Li et al., 2024). CertaIndex uses the same structure in its multi-path setting, measuring entropy over answers from independently sampled trajectories.

Probing a single chain of thought is a different object, and we call it *self-consensus* to keep the two apart. Its readings are not independent samples: every probe is conditioned on the same prefix, so agreement between them records that the prefix has not changed its answer, not that separate lines of reasoning have converged on one. Two consequences follow, and they organize Section 5. Agreement no longer supplies a vote that can outweigh an error, so it offers no protection when a stop commits to a wrong answer. And what agreement does report—that the trajectory keeps giving the same answer—is only informative if the trajectory has finished reasoning, which is exactly what a mid-

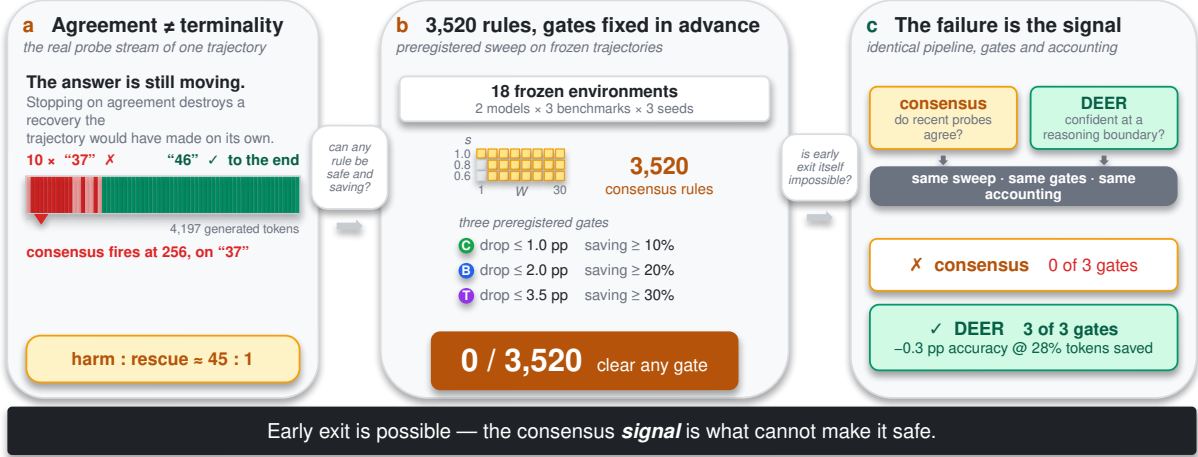


Figure 1: **The argument in one picture.** (a) Intermediate probe agreement is not terminal. On a real trajectory (MATH500 problem 68, DeepSeek-R1-Distill-Qwen-7B, dense 64-token probes) the model emits the same wrong answer, 37, for ten consecutive probes; a rule requiring three agreeing probes fires at generated token 256 and commits it. Left to run, the trajectory reaches the correct answer 46 and holds it for its final 47 probes. Ribbon colours: red the dominant wrong answer, pink other wrong answers, green correct. (b) We ask whether *any* consensus rule escapes this. A preregistered sweep of 3,520 rules—window size $W \times$ share threshold s , crossed with probe schedule, maturity, validity and certainty—is replayed on 18 frozen environments against three acceptance gates fixed before the sweep ran. None clears any gate, and the two ways to fail are symmetric: capping the accuracy drop at 1 pp permits 0.2% token saving, while demanding 10% saving costs 2.7 pp. (c) The failure is the *signal*, not early exit. Swept through the identical pipeline, gates and token accounting, a boundary-confidence method (DEER) clears all three gates while consensus clears none—and it does so on the held-out test split, at $4\times$ scale and on a different architecture as well (§4.5).

trajectory probe cannot establish. The trajectory is still correcting itself, an intrinsic process distinct from the prompted, post-hoc self-correction that Huang et al. (2024) find unreliable; we measure answer evolution *within* one uninterrupted trajectory, and find (Section 4) that a model’s mid-trajectory answer is far from final.

Alternative termination signals. Rather than probe agreement, a stop can read other signals: verifier- and process-reward-guided decoding (Lightman et al., 2024; Wang et al., 2024) score partial reasoning, and adaptive-computation methods learn a halting signal at the token or layer level (Graves, 2016; Schuster et al., 2022). DEER (Yang et al., 2025) stops on the model’s confidence in a trial answer at reasoning boundaries. We use DEER only as a *positive control* (Section 5): run through the identical pipeline and acceptance gates, it clears them where no consensus rule does, indicating that the failure is specific to the probe-agreement signal rather than to early exit itself. We do not claim this broader class of signals is categorically superior—our experiments isolate one signal, not a taxonomy of methods. Test-time scaling trades compute for accuracy via repeated sampling and search (Snell

et al., 2024; Brown et al., 2024); early exit is its dual, spending less without losing accuracy.

Preregistration. To keep an exhaustive rule search from silently overfitting to a favorable operating point, we fix the rule schema, splits, and acceptance gates before looking at held-out data (Nosek et al., 2025). This preregistration (Section 3) is what makes a *negative* result interpretable.

3 Method

To ask whether *any* probe-based rule can succeed, we need (i) an evaluation that a rule cannot game by changing the reasoning it observes, and (ii) a selection procedure that cannot silently overfit to a lucky operating point. We preregister both.

3.1 Probe-independent, frozen trajectories

For each problem we generate exactly *one* main trajectory in a single request (caps: 16K tokens for MATH500 and AMC23, 32K for AIME24) and freeze it. All probing is then performed *offline* against fixed text prefixes of that frozen trajectory. Because probes never re-enter the decode, changing a probe schedule—or a stopping rule—cannot

change the underlying reasoning; every rule is scored against the same trajectories. We build two offline probe banks: a *dense* bank that re-probes every 64 tokens with simple@32 (a 32-sample boxed-answer probe), and an *adaptive* bank that additionally fires event-triggered probes at entropy drops and at conclusion/reflection/answer markers.

3.2 Token accounting

For a rule that stops a problem at token position s after consuming probe output p , we charge total decode tokens $T = s + p$ and compare against the frozen baseline B (the full trajectory, capped at budget). *Net* savings is $(B - T)/B$ and *gross* savings is $(B - s)/B$; their gap is the probe cost. The accuracy drop compares the correctness of the committed answer at s against that of the frozen final answer (it is not an agreement rate between them). Two quantities therefore move independently, and we track both: *where* the rule stops (setting the accuracy drop and the main-token saving s) and *how much it probes* to get there (adding p). Section 5 shows that only the first is intrinsic.

3.3 The consensus rule space (W, s)

The consensus signal any probe-based rule can read reduces to two hyperparameters: a **window size** W (how many of the most recent probes to inspect) and a **share threshold** s (what fraction of that window must agree on a single answer before a stop is allowed). This pair spans the family to a close approximation: $W=1$ is the “latest-probe” rule, $s=1.0$ is strict unanimity over the last W probes (the CertIndex-style stop), and an entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid. We sweep $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$ and $s \in \{0.6, 0.8, 1.0\}$, deliberately including *large* windows so a rule can demand a long, stable agreement before committing.

Around this core we vary only the operational knobs that govern *when* and *how* the signal is read: **probe schedule** (fixed interval $\in \{64, 128, 256, 512\}$ tokens, or event-triggered probing at conclusion/entropy/reflection markers); **maturity** (a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed); **validity** (an answer-shape filter, rejecting empty/letter answers or only empties); and **certainity** (optionally requiring the supporting probes to be flagged certain). Enumerating this prereg-

istered grid over a fixed-schedule and an event-triggered family yields 3,520 candidate consensus rules; formal definitions are in Appendix A.

A better signal, swept identically. To test whether the failure is early exit *itself* or the consensus *signal*, we sweep one non-consensus method through the same pipeline, gates, and token accounting. **DEER** (Yang et al., 2025) stops on the model’s own confidence in a trial answer at a reasoning boundary; its only hyperparameter is a confidence threshold, which we sweep over 14 values. We use the stronger *trial-answer-submit* variant (commit the trial answer directly, no separate readout) rather than the faithful readout reproduction—our aim here is not to reproduce prior work but to ask whether *any* rule can clear the gates. Placing DEER and consensus on one frontier under identical accounting is what makes the comparison a controlled test of the signal.

3.4 Models, benchmarks, splits

Development uses two models—DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B—on three benchmarks (MATH500, AMC23, AIME24), each at seeds 42/43/44: 18 model $\times$ benchmark $\times$ seed environments. Problems are partitioned *by problem id* into 60/20/20 train/dev/test splits (MATH500 300/100/100, AMC23 24/8/8, AIME24 18/6/6), so no problem leaks across splits. Confirmation reserves seeds 45/46/47 plus two unseen models—DeepSeek-R1-Distill-Llama-8B and Distill-Qwen-32B—all evaluated on the *test* split only. In one line: *train* generates and screens candidate rules in-sample, *dev* is the sole selection target the gates are read on, and *test* (with the two unseen models) is read once for confirmation after the rules and thresholds are frozen—never for tuning. Where later sections say “train+dev,” they mean a rule was required to pass a gate in-sample on train and is then measured on dev.

3.5 Preregistered operating points and gates

We fix three operating points and their acceptance gates *before* looking at held-out data (Table 1). Each gate is applied in order: first the **total accuracy drop** (macro-averaged over the 18 environments) must stay under a cap; then the **total net token saving** (macro-averaged) must clear a floor; and finally a **positive-saving fraction** (psf, the fraction of environments with strictly positive net saving) floor must be met. A rule is accepted

Operating point	Max total acc. drop	Min total saving	psf floor
conservative	1.0 pp	10%	0.80
balanced	2.0 pp	20%	0.80
token_efficient	3.5 pp	30%	0.70

Table 1: Preregistered acceptance gates. A rule must keep the macro-averaged accuracy drop under “max total acc. drop,” deliver at least “min total saving” net tokens, and post positive net saving on at least a psf fraction of environments. Gates are fixed in advance and *not* relaxed post hoc.

at an operating point only if it satisfies all three on train+dev. The saving floor is what distinguishes a genuinely useful early exit from a rule that stays accurate only by almost never stopping.

3.6 Metric resolution

The total accuracy drop is a macro-average over the 18 environments (each model×benchmark×seed weighted equally). We span two models, three benchmarks, and three seeds and weight them equally so that selection rewards a rule that is robust across benchmarks, models, and randomness, not one that overfits a single setting. Pooling problems instead would let MATH500 (100 dev problems per seed vs. 8 for AMC23 and 6 for AIME24) dominate the headline, so a rule that wins only on MATH500 could pass; equal per-environment weight forces a selected rule to hold on the harder competition sets too. Macro also keeps any single problem from swinging the result—one AIME24 problem is 16.7 pp of a 6-problem split but only $\sim 1/108$ of the macro drop—while the three seeds damp the extra per-cell noise of the small sets. The gates are read on the dev split (the selection target) with train reported alongside; because *no* consensus rule clears any gate by a wide margin (Section 4), the finding does not rest on a single fragile cell. Held-out confirmation on the test split and two unseen models provides the complementary out-of-distribution check.

3.7 Preregistration commitments

Three commitments make the result interpretable. (C1) The test split and both confirmation models are never read during rule selection. (C2) All three operating points and their gates in Table 1 are fixed before seeing held-out data and are not relaxed afterwards; if no rule passes a gate, that is reported as the finding rather than engineered away, and all three points are reported regardless of outcome.

(C3) The frozen rule set is fixed on train+dev and its hash recorded before any test/confirmation evaluation. Section 4 reports what happened when we applied C2.

4 Experiments

Section 5 argues that probe agreement cannot make a stop safe on its own, because the probes are not independent readings. The experiments here establish that this is not a matter of tuning: we search the rule space exhaustively and under commitment, and find nothing that escapes it.

We first confirm the phenomenon the paper turns on—intermediate consensus is non-terminal (§4.1)—then report the preregistered sweep and its central negative result: *no* consensus rule is both safe and saving (§4.2). We then show the failure is specific to the *signal*, not to early exit, by sweeping a non-consensus method (DEER) through the identical pipeline and gates and finding that it *does* clear them (§4.3); a released consensus stopper, CertIndex (CoT), shows the same failure at its default setting (§4.4). Finally we confirm the frontier out of distribution (§4.5).

4.1 Non-Terminal Agreement

We log a dense probe stream for every MATH500 problem under DeepSeek-R1-Distill-Qwen-7B at the main 16K/32K budgets (all 500 problems, three seeds; 1,500 trajectories), never intervening—the probe is a separate forward pass that only observes the main trajectory (full setup in Appendix G)—and read off four facts.

(1) **Final agreement is reliable; early agreement is not.** By the time a trajectory finishes, agreement does track correctness: *cumulative* agreement across all probes is 97.8% correct (186 trajectories reach it), and even the final five-probe *window* is 90.4% correct (1,205). But an online rule cannot wait for the end—it fires the *first* time probes agree, mid-trajectory, where the answer is far from settled (facts 2–4). The operative question is not whether *final* consensus is trustworthy, but whether *early* consensus is—and it is not. (2) **Recovery is common.** Of the 1,148 trajectories whose *first* probe is wrong, 89.1% eventually flip to correct; of the 736 that formed a three-probe consensus *different* from their final answer, 84.2% ended up correct. (3) **Later consensus is worse.** Consensus formed before 512 tokens is 91.0% correct, but consensus forming only after 2048 tokens is 71.6%

correct—hard problems converge slowly *and* to worse answers. **(4) The cost lands at the stop.** A naive stop (first three consecutive agreeing, certain, non-empty probes) fires on 1,477/1,500 trajectories and commits to answers correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%: a 40.2 pp loss.²

Figure 2 shows where this cost is concentrated. Expressing the moment consensus first forms as a *fraction* of each trajectory’s own length—which controls for problem difficulty—half of all consensus forms within the first tenth of the trajectory, and there the answer they commit to is correct only 27.5% of the time against 85.2% for the same problems run to completion. The gap narrows as consensus forms later, but an online rule cannot choose to wait: it fires when agreement first appears, which is precisely where agreement is least informative.

A good stopping rule must therefore be much more than “agreement.” The rest of this section asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical.

4.2 Sweep Results

We replay all 3,520 consensus rules on the frozen trajectories and aggregate to per-environment metrics—126,720 rows = 3,520 rules $\times$ 18 environments $\times$ 2 splits (train, dev)—then, per rule, its total (macro) accuracy drop and net token saving (accounting of Section 3.2). Applying the preregistered gates yields the negative result that Section 5 explains.

The outcome holds across environments: across all 126,720 rule–environment evaluations accuracy strictly *falls* in 64.9% and strictly *rises* in only 10.5% (unchanged in 24.6%), for a mean drop of 13.0 pp; over rules the total dev drop has median 13.2 pp and a minimum of 0.81 pp. **Not one of the 3,520 rules clears any gate.** The reason is the shape of the trade-off, which Figure 3 reads directly off the sweep’s own two axes: staying accurate and saving tokens are in direct tension. Only 40 rules keep the total drop at or below the conservative 1.0 pp cap, and the most any of them saves is 0.2%—far under the 10% floor; conversely, the

²A simplified heuristic, *not* the CertIndex (CoT) rule (§4.4); it fires early because the first stable, certain agreement is typically a mid-trajectory placeholder. The swept rules below and the mechanism of Section 5 quantify the same directional accuracy tax across the full rule space.

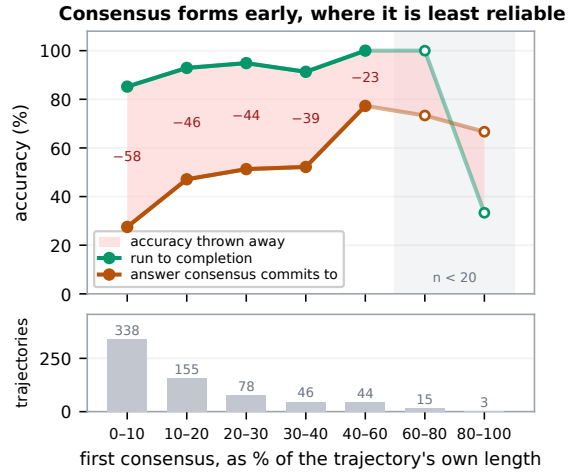


Figure 2: **Consensus forms early, where it is least reliable.** For each dev-split trajectory we take the first window of three agreeing non-empty probes and ask two things: is that answer correct, and would the same problem have been correct if left to run? Position is a fraction of the trajectory’s *own* length, so 500 tokens into a 600-token trajectory counts as late and 500 into a 16K one does not. Consensus forms in 679 of 684 trajectories, half within the first tenth; the shaded gap is the accuracy a stop at that moment throws away. Open markers mark bins with fewer than 20 trajectories; the final bin ($n=3$) should not be read.

first rule to save 10% already costs 2.66 pp, saving 20% costs 6.17 pp, and saving 30% costs 11.8 pp. Crucially, the sweep now includes *large* windows (up to $W=30$): these do buy low accuracy drop, but only by stopping so late that net saving collapses to zero. No setting of the consensus signal escapes the trade-off. Figure 3 makes this a property of the whole surface rather than of the frontier’s edge: read cell by cell, the safest rule available at each (W, s) approaches the 1 pp cap only as the saving it buys approaches zero, while every cell that saves substantially costs several points. No cell clears the conservative gate.

4.3 A Non-Consensus Signal

The same gates are cleared, comfortably, by a rule that does *not* read probe agreement. Sweeping DEER’s confidence threshold through the identical pipeline and token accounting (Section 3.3), its trial-answer-submit variant passes all three operating points on dev: the conservative point loses only 0.33 pp while saving 28.2%, the balanced point 1.03 pp at 29.6%, and the token-efficient point 2.75 pp at 31.9%; at a stricter threshold it is accuracy-neutral (−0.06 pp) while still saving

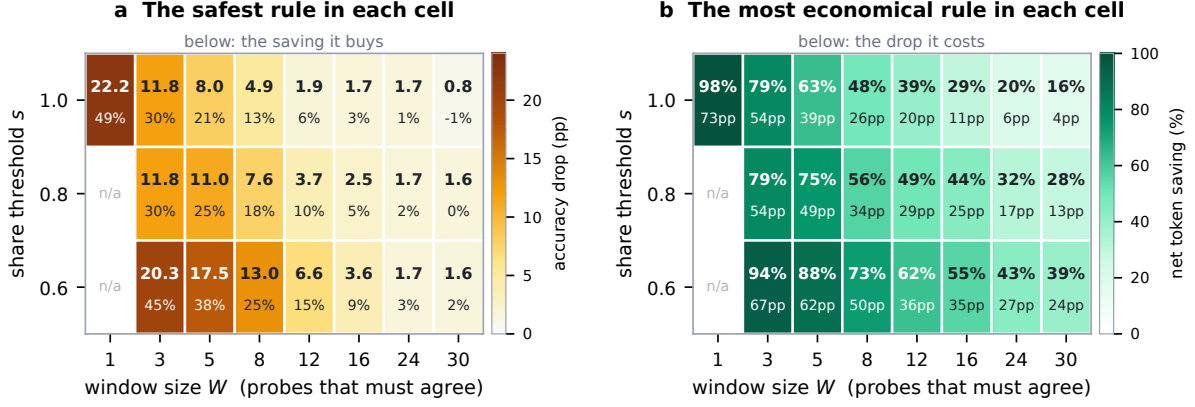


Figure 3: **The safe-and-saving corner is empty across the entire rule space.** Dev split, macro-averaged over the 18 environments. Each cell holds the 160 rules that share its (W, s) and differ only in the operational knobs, summarised by the two members a gate interrogates. **(a)** the cell’s *safest* rule: colour is its accuracy drop and the number beneath is the saving that rule buys. Larger windows and stricter thresholds do drive the drop towards the 1.0 pp cap—but the saving falls with it, reaching -1% at $W=30, s=1.0$. **(b)** the cell’s *most economical* rule: colour is its net saving and the number beneath is the drop it costs. Substantial saving is available everywhere, at 4–73 pp. A cell would be outlined in green if any of its rules cleared the conservative gate (drop ≤ 1.0 pp and saving $\geq 10\%$); none is.

Method	Acc.	Δacc (macro)	Net saving
Full generation	82.5%	—	0%
Consensus (drop ≤ 1.0)	81.6%	-0.9 pp	$+0.2\%$
Consensus (save $\geq 10\%$)	79.9%	-2.7 pp	$+10.7\%$
DEER (conservative)	82.5%	-0.33 pp	$+28.2\%$
DEER (balanced)	81.9%	-1.03 pp	$+29.6\%$
DEER (token_eff.)	80.1%	-2.75 pp	$+31.9\%$

Table 2: Dev operating points (macro over 18 environments). No consensus rule is simultaneously safe (≤ 1.0 pp) and saving ($\geq 10\%$): capping the drop forces saving to near zero, and demanding saving forces a large drop. DEER, reading a non-consensus signal, clears all three gates on the same trajectories under the same accounting.

20.8%. Early exit is therefore not impossible; the consensus *signal* is what cannot make it safe (Table 2).

4.4 CertIndex (CoT) at Its Default

The sweep covers rules we constructed; a released method lets us see where a default setting lands. We reproduce two prior stoppers in a common harness—the same frozen trajectories, the same token accounting. **CertaIndex (CoT)** (Fu et al., 2024) is the consensus stop for a single chain of thought, reproduced from the released implementation at its default `mid` setting: stop once three consecutive probes give the same non-empty an-

swer and none hedges. We use the authors’ probe prompt and probe schedule, so this is their rule as shipped, not a variant of ours. **DEER** (Yang et al., 2025) here is the readout reproduction (the stronger trial-answer-submit variant is what we swept in §4.3).

Two scope notes. First, CertIndex’s multi-path setting is ordinary self-consistency: it measures entropy over answers from *independently sampled* trajectories, and nothing here bears on it. What we reproduce is the single-trajectory case, where the same statistic is read over repeated probes of one chain. Second, Table 3 reports these reproductions on our frozen trajectories and benchmarks; it is not a re-run of either paper’s end-to-end system, and the original CoT results were reported on an easier benchmark, where intermediate answers settle sooner.

At a 99.8% stop rate and 70 pp lost, CertIndex (CoT) lands where the swept consensus frontier says a rule at that stop rate must land. This is a property of the signal, not a defect of the implementation: it is the same trade-off §4.2 maps across the whole rule space, seen at one shipped operating point. The DEER readout, reading a non-consensus signal on the same trajectories, stays near full accuracy; the stronger trial-answer-submit variant we swept turns that into a gate-clearing frontier (§4.3). That only the non-consensus signal survives is what we analyze in Section 5.

Method	Model	Accuracy	Δacc	Fair token saving	Stop rate	Signal
Full generation	Qwen3-8B	85.4%	—	0%	0%	—
Full generation	DeepSeek-R1-Distill-Qwen-7B	79.8%	—	0%	0%	—
CertaIndex (CoT)	Qwen3-8B	15.3%	−70.1 pp	90.1%	99.8%	consensus
CertaIndex (CoT)	DeepSeek-R1-Distill-Qwen-7B	23.9%	−55.9 pp	76.7%	98.7%	consensus
DEER	Qwen3-8B	86.2%	+0.8 pp	16.3%	41.4%	boundary conf.
DEER	DeepSeek-R1-Distill-Qwen-7B	74.9%	−4.8 pp	20.2%	56.1%	boundary conf.

Table 3: Prior early-stoppers reproduced on our frozen trajectories (dev, macro over three benchmarks; probe tokens charged). **CertaIndex (CoT)** stops on almost every problem and stops early, and loses 56–70 pp while saving 77–90% of tokens—the pattern §4.1 describes, at a single shipped operating point. **DEER**, reading boundary confidence rather than probe agreement, stays near full accuracy (+0.8 pp on Qwen3-8B) while still saving tokens. These are reproductions on our benchmarks, not the original papers’ end-to-end numbers; Section 5 interprets the contrast.

4.5 Generalization

We read the test split once (commitment C3). Because the development result is that *no* consensus rule clears any gate, there is nothing to confirm by re-selection; we ask whether the *frontier* reproduces, ranking rules on dev and measuring on test.

Held-out test split. Each consensus rule’s total drop on test tracks its dev value closely ($r = 0.98$). The split-invariant claim is the empty *joint* gate: **no consensus rule clears the conservative gate on both splits**—0 on dev, and the 444 rules admissible on *test alone* are exactly the in-sample winners the held-out split exists to reject (all fail on dev). DEER, by contrast, remains admissible across splits: 3 of its thresholds clear the conservative gate on *both* dev and test (and 5 clear balanced on both), so the cross-signal contrast is not a dev artifact.

Unseen scale (4×) and architecture. Both unseen models are evaluated on the test split at three seeds (45/46/47; nine environments each), so their gates are as well resolved as development’s. On Distill-Qwen-32B, never used in development, the consensus frontier reproduces ($r = 0.97$ against dev drop); on the held-out architecture and family DeepSeek-R1-Distill-Llama-8B it reproduces at $r = 0.94$.³ The *conservative* gate stays empty on every model—dev, 32B, and Llama all admit 0 rules (the best rule under a 1.0 pp drop saves 0.6% on 32B and 9.3% on Llama). At the looser gates a scale effect appears: on the 4× larger 32B a few consensus rules become admissible in-sample (4 at

balanced, 6 at token-efficient), reflecting that a bigger model’s intermediate answers stabilize somewhat earlier—but these are not the dev-selected rules (dev admits none), and on the same-scale Llama architecture the gates stay empty (0/0/0). Consensus thus yields no rule that is safe-and-saving where it is selected, on any model. DEER, swept identically on these unseen models, **clears the gates on both**: on Qwen-32B it is accuracy-neutral at the conservative point (−0.24 pp while saving 32.4%, $\tau=0.97$) and on Llama-8B it loses 0.67 pp at 26.7% ($\tau=0.99$), passing all three gates. Consensus is unsafe, and DEER safe-and-saving, across all four axes we vary—benchmark, seed, 4× scale, and architecture/family.

Figure 4 shows the selection pipeline end to end. The 478 consensus rules that clear the conservative gate in-sample on train are tracked onto dev and test as a band; the three dev-selected DEER thresholds are tracked individually. On dev—the split the gate is applied on—not one of the 478 is admissible, and the joint gate is empty, while all three DEER points stay inside on every split. The dev and test splits are not equally demanding: the same 478 rules have a median drop of 4.5 pp on dev but 0.6 pp on test, an asymmetry driven by the two smallest environments (AMC23 and AIME24 under Qwen3-8B, 8 and 6 problems), which carry the same weight as a 100-problem environment under macro-averaging. This is why the claim is stated jointly rather than per split: 364 of the 478 are admissible on test, but they are exactly the in-sample winners a held-out split exists to reject, and none of them is selectable from dev.

Per-model and per-benchmark views of the same data, together with an *oracle* upper bound—a perfect probe-based stop committing the earliest cor-

³The Llama-8B trajectories were re-collected after correcting a chat-template defect (a missing model-specific beginning-of-sequence token that degraded its generations); the corrected model scores $\sim 88\%$ on MATH500.

rect probe, which sits at a negative drop of 2–5 pp at 40–80% saving, far in a corner neither method reaches—are given in Appendix E.

5 Analysis

Section 4 shows that no rule in the searched space is both safe and saving. This section gives the reason, which is what makes the result more than a report on 3,520 particular rules. The argument has three parts. Probe agreement lacks the independence that makes agreement informative in the first place (§5.1), so it cannot protect a stop from being wrong. What it does report is often a forced placeholder rather than a settled answer (§5.3). And widening the window, the obvious remedy, works only by declining to stop (§5.4). The accuracy cost that follows is a property of the reasoning trajectory, not of a threshold or of probe density.

5.1 The Role of Independence

Consensus early exit does not claim to *improve* accuracy; its goal is to save tokens without losing any. The question is therefore not whether agreement adds information, but whether it provides enough protection to make stopping safe. It does not, and the reason is structural.

Self-consistency’s majority vote can absorb an error because its paths are sampled independently (§2): a wrong answer on one path is outvoted by others that reasoned differently. Probing one trajectory yields no such thing. Every probe reads the same prefix, so the readings move together; when the prefix is wrong, they are wrong together, and a majority over them cannot correct anything. Agreement across probes is a statement about the stability of one line of reasoning, not corroboration from several.

This is why the accuracy cost is so large when it appears. A consensus rule has no mechanism that recovers from a bad stop: whatever the probes agree on is committed, and the vote that would have caught the error in the multi-path setting does not exist here. The rest of this section shows what the probes actually agree on, and why the standard remedy does not repair it.

5.2 Probe Wording versus Position

If early agreement reflected a settled belief, the answer a probe returns should not depend much on how the probe is worded. We test this directly on paired re-probes of the *same frozen prefixes*: at

Position (% of trajectory)	<i>n</i>	Wordings agree	Probe correct
0–10	1,025	42.7%	17.9%
10–20	1,184	52.2%	27.7%
20–40	1,696	63.6%	46.1%
40–70	1,669	82.3%	71.5%
70–100	951	88.9%	79.7%

Table 4: Two probe wordings applied to the *same* frozen prefix (DeepSeek-R1-Distill-Qwen-7B, MATH500). Early in a trajectory the two wordings disagree more often than they agree; by the end they nearly always coincide. Position is a fraction of each trajectory’s own length.

every probe position we read the trajectory with two different probe wordings (our boxed-answer suffix and the CertIndex suffix, both capped at 32 tokens), so the model’s state is identical and only the elicitation differs.

Table 4 shows the result. In the first tenth of a trajectory the two wordings return different answers 57% of the time; by the final third they disagree only 11% of the time. Early “answers” are therefore substantially a property of the question asked rather than of the state being read, and they become properties of the state only as the trajectory finishes.

This distinguishes our account from the obvious alternative. If the model simply held a stable but mistaken belief early on, both wordings would read back the *same* wrong value; instead they disagree. What a consensus rule treats as a converging answer is, early on, an output the probe has elicited. Note also that probe wording has almost no effect on *final* accuracy in the same paired data (a 0.65 pp readout effect against a 9.15 pp effect of *when* one probes), so the sensitivity above is specific to early positions and is not a defect of any particular suffix.

5.3 Error Taxonomy

Two annotators independently labelled all 134 stopped-but-wrong cases (Figure 5, $\kappa = 0.82$): only one in five is a case where the model had genuinely settled on a wrong value. Three in five are answers it had not converged on at all, and a further one in six are artifacts of the probe’s output format.

The reason is that a probe does not read the model’s belief; it *forces* an answer. A model that has not finished reasoning must still write something in the box, and the cheapest thing to write is what it wrote last time. Repetition of a forced answer is therefore stable for reasons that have

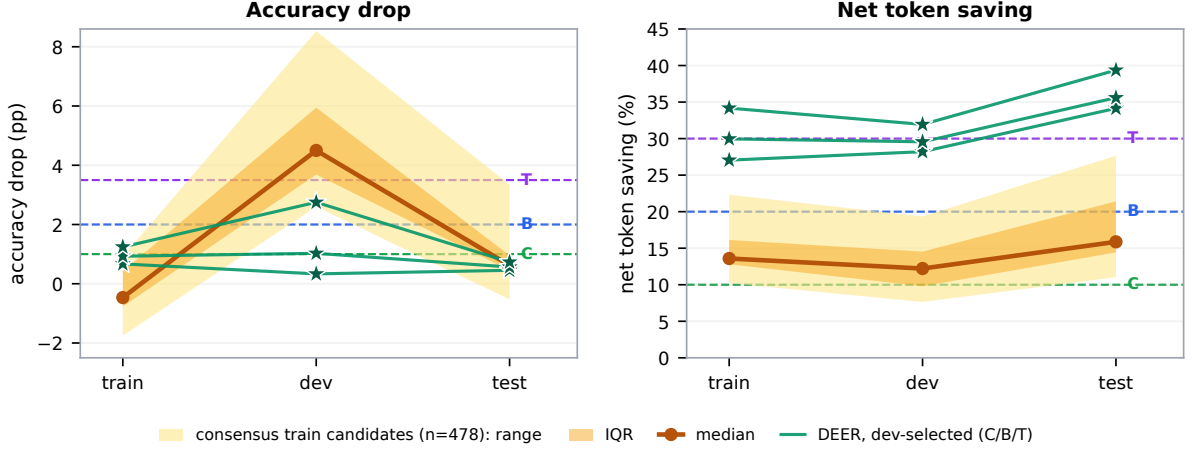


Figure 4: **A rule must clear the gate on every split.** The 478 amber rules are exactly those that clear the conservative gate *in-sample on train*; drawing all 478 would be unreadable, so we show their median, interquartile range and full range. Dashed lines mark all three preregistered gates (Conservative, **B**alanced, **T**oken-efficient); a rule must sit below them on the left and above them on the right. On dev, the split the gate is applied on, none of the 478 does. Some recover on test, but a rule selected on dev is never among them: jointly, 0 of 478 clear all three splits, against 3 of 3 for the DEER operating points selected on dev. Macro-averaged over 18 environments per split; train and dev share seeds 42–44 and differ only in problems, while test uses unseen problems and seeds 45–47.

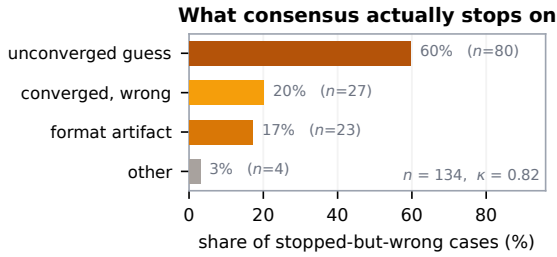


Figure 5: **What consensus actually stops on.** All 134 stopped-but-wrong cases, labelled independently by two annotators ($\kappa = 0.82$ on the coarse coding). Fewer than one stop in five is the model being wrong about something it had actually decided. The cases were collected under a *short* window (three consecutive agreeing certain probes), which is precisely the regime in which unconverged answers dominate—consistent with Figure 6, where enlarging the window buys accuracy back only by declining to fire.

nothing to do with the reasoning being finished, and a consensus rule cannot tell that case apart from a settled one. This also predicts the window effect of §5.4: a longer window is a longer buffer in which a placeholder can be overturned, which is why it reduces false consensus without removing the underlying risk.

5.4 Effect of Window Size

The natural remedy is to demand more agreement before stopping. It helps, but only by stopping

less often. Consider only the stops where committing early *changes* correctness relative to the trajectory’s own final answer, and count two outcomes: (final-correct, stop-wrong)—a correct recovery destroyed—versus (final-wrong, stop-correct)—a wrong answer luckily banked. Pure sampling noise predicts a $\approx 1:1$ ratio; a genuine “continued reasoning corrects” effect predicts $\gg 1$. On the frozen replay this ratio depends strongly on the window: it falls from $\approx 45:1$ for a latest-probe stop ($W=1$) to $\approx 2:1$ at the largest window ($W=30$). But the low end is bought only by stopping so rarely that net saving vanishes ($W=30$ fires on 121 dev problems vs. 668 at $W=1$); at any window that actually saves tokens the ratio stays well above 1:1. DEER, by contrast, holds $\approx 2.4\text{--}3.5:1$ at its three operating points *while* saving 28–32%, because its boundary-confidence signal commits mainly once the trajectory has settled. Stopping on consensus therefore removes a correct-in-the-end answer many times more often than it rescues a wrong one—directional, not sampling noise. This is the recovery phenomenon of §4.1 (84.2%, 89.1%), now at the main 16K/32K budgets and as a signed effect. Figure 6 plots the whole window axis, so the claim rests on the shape of the curve rather than on either endpoint: the ratio and the saving fall together, and DEER sits at consensus’s best ratio and its best saving simultaneously, which no window does.

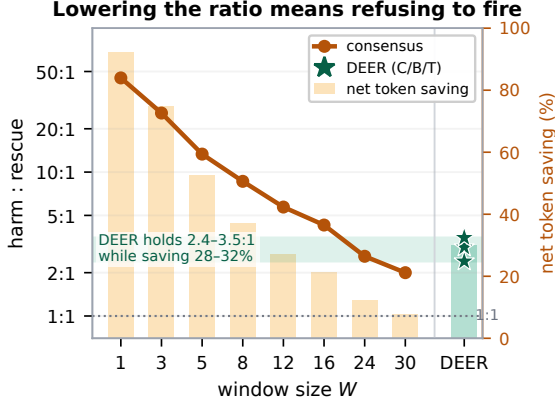


Figure 6: **Lowering the ratio means refusing to fire.** Among stops whose committed answer differs in correctness from the trajectory’s frozen final answer, we count recoveries destroyed against wrong answers banked; sampling noise predicts 1:1. Widening the window brings the ratio down from 45:1 to 2:1, but only by stopping on ever fewer problems (668 $\rightarrow$ 121 of 684), so net saving collapses from 92% to 8% alongside it. Dev split, 18 environments; one consensus rule per W (share threshold 1.0, probe interval 128, maturity floor 512, schema validity), with W the only axis that varies.

5.5 Case Studies

The three regimes above appear together on individual trajectories (DeepSeek-R1-Distill-Qwen-7B, MATH500, probes every 64 tokens; $a \times n$ denotes an answer repeated over n consecutive probes). Appendix F gives the full streams and discussion.

On problem 320 the model answers 0 at its *first* probe and holds it for 27 probes—nearly half a 3,683-token trajectory—before reasoning its way to the correct $-\frac{24}{25}$. The committed value is not a considered wrong answer but one emitted before any real work, and its stability is correspondingly uninformative. The run is also long enough to defeat every window we searched: even $W=24$ fires here and commits 0, which is why widening the window is not a remedy for a placeholder that outlasts it.

Problem 253 asks what fraction of two feet is three inches, a free-response question with no options. Every probe from token 384 to token 1600 returns the letter D; the correct $\frac{1}{8}$ appears only at the final probe. Agreement is total and refers to nothing, which is the format artifact of §5.3 in its clearest form.

The minority case is worth stating too. On problem 240 a three-probe rule commits 52 while the trajectory itself ends at 154; the gold answer is

116, so both are wrong and stopping destroys nothing. This is the regime consensus early exit was designed for—the model was not going to solve the problem, so a stop is free. The difficulty is that agreement cannot tell this case apart from problem 320, and problem 320 is the more common one.

5.6 Probe-Independence of the Accuracy Cost

Why is the ratio structural? For a rule stopping at token position s , the accuracy drop depends only on s : committing at s forgoes whatever correction the trajectory would have made after it. It is a property of the reasoning, not of the probe, so it survives *any* probing scheme. This is what never reaches zero in the sweep. The probe grid (intervals 64/128/256/512 tokens) makes essentially every stop position reachable and the observed floor is the minimum over these schedules; a sparser schedule can only stop later or at a subset of positions, never at a safer one the grid missed. Hence no threshold on the consensus signal *within the searched space* escapes the accuracy tax; the searched space is a lower bound on the cost, not a lucky miss. This does not turn a finite rule search into a universal impossibility theorem—it motivates caution for consensus variants beyond the preregistered schema rather than proving a universal claim about them. In short, agreement reports what the model has said recently, not whether its reasoning has finished.

5.7 Accuracy Tax versus Probe Tax

The natural rebuttal is that the negative result is an artifact of charging dense re-probing. It is not, and separating two costs shows why. Besides the accuracy tax, a rule pays a *probe tax*: the probe output p (Section 3.2). Under dense re-probing a late-stopping large-window rule pays p over a long prefix, opening a gap between its *gross* savings and its (smaller) *net* savings (Table 5). Unlike the accuracy tax, the probe tax scales with probe density and is reducible (sparser or shorter probes, KV-cache reuse). The two are separable: the *savings* axis is partly a probe-tax artifact and is addressable, but the *accuracy* floor is probe-independent and is what makes the corner empty—and it is the accuracy floor, not the probe tax, that DEER escapes.

5.8 Locating the Failure

The two experiments of §4.3–4.4 localize the cause. Every method that stops on *probe agreement*—the swept consensus family and the CertIndex

Consensus point	Gross saving ($B - s$)/ B	Net saving ($B - s - p$)/ B
save $\geq 10\%$ (drop 2.7 pp)	+14.7%	+10.7%
save $\geq 20\%$ (drop 6.2 pp)	+27.0%	+20.2%

Table 5: Gross vs. net savings for two consensus frontier points; the gap (~ 4 – 7 pp) is the probe tax. It shifts the *savings* axis but never rescues the accuracy drop—which is the intrinsic tax that keeps the corner empty.

(CoT) reproduction alike—fails a gate or loses many points; the one method that clears the gates, DEER, reads not probe agreement but the model’s own confidence in a trial answer at a reasoning boundary. Because DEER is swept through the *identical* pipeline, gates, and token accounting, this is a controlled test: the same early-exit machinery fails on the consensus signal and succeeds on a non-consensus one, so the failure is a property of the *signal*, not of early exit.

The constructive reading follows directly—a safe signal should track whether the *trajectory itself* will still change, not what recent probes agree on; verifier- and process-reward-based termination (Lightman et al., 2024; Wang et al., 2024) and learned halting (Graves, 2016; Schuster et al., 2022) point the same way. Turning any of these into a *confirmed* method—with its own train/dev $\rightarrow$ test validation—is the natural next step; our contribution is to establish rigorously which signal not to build it on *by itself*.

That qualification matters. Our claim is about consensus used as the sole criterion for stopping, which is the use we searched and the use that fails. Probe agreement is cheap to compute and does carry one piece of information—that the current answer has not changed—and combined with a signal that reflects the model’s own estimate it may well be useful; CertaIndex composes multiple indicators for related reasons (Fu et al., 2024). Nothing here argues against such combinations. What the sweep rules out is the step of treating agreement, on its own, as sufficient evidence that reasoning has finished.

Two further caveats bound the claim: it concerns *consensus-based* rules within the searched schema and dense-probe regime, and it does not say early exit is impossible—only that this signal cannot make it safe alone.

6 Conclusion

We asked whether intermediate answer-consensus is a safe signal on which to early-exit a reasoning LLM. Used on its own it is not, and not because the rule is untuned. Agreement is informative in self-consistency because the paths are sampled independently; probes of one trajectory share a prefix, so they cannot outvote each other’s errors. What they agree on is often an answer the model has not settled on: three in five of our hand-labelled wrong stops commit to a placeholder the probe forced it to produce. Stopping on such an answer destroys a correct-in-the-end answer far more often than it rescues a wrong one (up to $\sim 45\times$ at an aggressive latest-probe stop, $\sim 2\times$ at the largest windows, where the rule has nearly stopped firing). This accuracy tax is a property of the trajectory rather than the probe, and it shows up as an empty safe-and-saving corner across a preregistered space of 3,520 rules—not one of which clears any gate—reproduced on a held-out split and two unseen models, and visible at one shipped operating point in CertaIndex (CoT), which loses 56–70 pp. That the corner is empty for consensus but not for early exit is the point: swept through the identical protocol, a method reading the model’s own boundary confidence (DEER) clears all three gates on the very same trajectories, so the failure is specific to consensus. The takeaway for anyone building safe early exit is therefore about the choice of signal: read whether the trajectory itself has settled, not whether its recent probes agree. This does not put probe agreement beyond use—it is cheap, and it does report that the current answer has not changed. What our results rule out is treating that report, by itself, as evidence that the reasoning is done.

Limitations

The mechanism is argued, not isolated experimentally. Our account of *why* consensus fails—that probes of one trajectory are not independent readings, and that they frequently record a forced placeholder—rests on a structural argument (Section 2) together with the hand-labelled error taxonomy and the window sweep. We do not run a controlled experiment that manipulates independence directly, and the labelled cases were collected under a short window, the regime where unconverged answers are most common. A stronger test would construct probe streams with varying degrees of dependence and measure how the accuracy tax re-

sponds.

Scope of the negative result. The central result—no consensus rule clears any gate—is established on 18 development environments (two models, three benchmarks, seeds 42/43/44) and confirmed on the held-out test split and two unseen models (Section 4). It is a statement about the space searched: one boxed-answer probe suffix (simple@32) and one preregistered consensus schema (3,520 rules over a window-size $\times$ share-threshold family plus operational knobs). Other probe wordings and signals outside this schema are not searched; the mechanism (Section 5) is our argument that consensus variants would not help, but it is an argument, not an exhaustive proof.

DEER as the positive control, not a benchmarked SOTA. We sweep DEER’s trial-answer-submit variant to establish that *some* signal clears the gates the consensus family fails (on the development models, the held-out test split, and both unseen models); we do not claim it is the best possible early-exit method. Its role is a controlled positive control for the signal, not a leaderboard entry.

Small held-out sets on hard benchmarks. AIME24 and AMC23 are small (dev splits of 6 and 8 problems per seed), and the unseen models contribute a single seed, so those per-environment cells are noisy. The total drop macro-averages over the 18 development environments (three seeds per benchmark $\times$ model), and we read the single-seed unseen models as frontier-reproduction evidence rather than as independent gate tests (Section 3.6).

Probe-tax dependence of the savings axis. Net-savings numbers charge dense re-probing (every 64 tokens, 32-token probes); a sparser or KV-cache-reusing schedule would change the *savings* axis and could move some rules to positive net savings. We separate gross from net savings (Section 5) precisely so the probe-tax-dependent part is distinguishable; the accuracy-tax result does not depend on the schedule.

Domain. All benchmarks are competition mathematics with checkable final answers. Whether false consensus and the accuracy tax transfer to open-ended reasoning, code, or agentic tasks is out of scope.

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A Rule Schema Details

The consensus schema of Section 3 centers on two hyperparameters and a few operational knobs, enumerating to 3,520 candidate rules over a fixed-schedule and an event-triggered family. Each rule fixes:

1. **Window size** $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$: the number of most recent probes inspected. $W=1$ is the latest-probe rule.
2. **Share threshold** $s \in \{0.6, 0.8, 1.0\}$: the fraction of the last W probes that must agree on one answer (share measured over the full window, so an empty or dissenting probe counts against consensus); $s=1.0$ is strict unanimity. An entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid, so it is not swept separately.
3. **Probe schedule**: fixed interval $\in \{64, 128, 256, 512\}$ tokens, or

`adaptive_event` with entropy-drop and marker triggers and a fallback interval.

4. **Maturity**: a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed.
5. **Certainty**: optionally require the supporting probes flagged certain (minimum fraction).
6. **Validity**: answer-shape filter (reject empty / single-letter answers to suppress the Type-E probe artifact of Appendix G), or accept any non-empty answer.

Certainty. A probe is *certain* when its boxed answer is emitted with high token-level confidence: concretely, the mean probability of the answer tokens under the probe forward pass exceeds a fixed threshold (the Certaindex certainty flag). For the simple@32 bank this is aggregated over the 32 samples as the fraction agreeing on the modal answer. Dimension 5 gates a stop on the fraction of the supporting probes that are certain.

Entropy triggers. The `adaptive_event` schedule fires an extra probe when the token-level predictive entropy of the main decode drops by at least ΔH relative to a trailing baseline (an “the model just committed” signal), and at conclusion/reflection/answer marker tokens (e.g., “therefore”, “the answer is”), subject to a minimum inter-probe gap and a fallback interval.

B Preregistration Text

The protocol version, split manifest hash, candidate-rule generation script, and the three operating-point gates (Table 1) were fixed before any held-out evaluation. Commitments C1–C3 (Section 3) were recorded with the protocol. All three operating-point gates were fixed in advance; when the conservative gate proved empty we followed C2 and report the empty gate as the finding rather than relaxing it, and all three points are reported for both consensus and DEER regardless of outcome.

C Reproducibility

Frozen trajectories, offline probe banks, the sweep archive (126,720 dev metric rows plus the DEER threshold sweep), and the aggregation scripts are released. Table 6 and Figure 3 were reproduced to the reported precision directly from the released sweep

Threshold τ	Total drop (pp)	Net saving
0.9999	−0.06	20.8%
0.999	0.06	25.4%
0.995	0.33	28.2%
0.99	1.03	29.6%
0.97	2.75	31.9%
0.95	5.87	34.8%
0.90	11.43	42.6%

Table 6: DEER trial-answer-submit threshold frontier on dev (macro over 18 environments, same token accounting). Unlike consensus, low drop coincides with substantial saving—the conservative/balanced/token-efficient gates are all cleared around $\tau \in [0.97, 0.995]$.

archive. Grading uses a robust answer-equivalence check that tries the raw and normalized reference forms and both string and math-expression equality; per-problem correctness for the frozen baseline is recomputed at replay time rather than trusted from collection.

D DEER Threshold Frontier

Figure 3 traces the consensus trade-off along both sweep axes; read down the window axis, the minimum attainable accuracy drop falls monotonically (from 22 pp at $W=1$ to 0.8 pp at $W=30$) while the maximum net saving falls with it (from 98% to 39%), so the low-drop region is reachable only where saving has already collapsed. Table 6 gives the DEER threshold frontier that does reach the gate region.

E Per-Model and Per-Benchmark Frontiers

Figures 7–9 give the saving-versus-drop view of the same selection and generalization results summarised in §4.5, resolved per split, per model and per benchmark, with the oracle upper bound on each panel. They add no claim beyond the main text; they show the distributions the summaries are drawn from.

F Case Studies

The mechanism of Section 5 is easiest to see on individual trajectories. All three below are DeepSeek-R1-Distill-Qwen-7B on MATH500, seed 42, probed every 64 generated tokens; we write $a \times n$ for an answer repeated over n consecutive probes. A rule requiring three agreeing probes fires at the third element of the first such run.

Case 1: a placeholder that outlasts every window we searched. Problem 320 asks for $\cos \angle RPS$ given $\sin \angle RPQ = \frac{7}{25}$ (answer $-\frac{24}{25}$). The probe stream is

$$0 \times 27, B \times 8, -24/25, B \times 7, \\ E, B \times 6, -24/25 \times 7$$

The model answers 0 at the very first probe and repeats it for 27 consecutive probes, covering tokens 64–1728 of a 3,683-token trajectory. It then passes through a stretch of bare option letters before reaching $-\frac{24}{25}$ and holding it to the end; the trajectory is correct.

This single case makes three points. The stop is not a considered wrong answer but a value emitted before any real work has been done, and it is perfectly stable for nearly half the trajectory. Its stability therefore says nothing about whether reasoning has finished. And because the run is 27 probes long, it defeats *every* window in our sweep: even $W=24$, the second-largest we searched, fires here and commits 0. Enlarging the window does not address a placeholder that persists longer than the window.

Case 2: the probe’s format, not the model’s answer. Problem 253 is “What fraction of 2 feet is 3 inches?”—a free-response question with no answer options. The stream is

$$3, B, 3, 24, B, D \times 20, 1/8$$

From token 384 to token 1600, more than two thirds of the trajectory, every probe returns the letter D. There is nothing for D to refer to. The model is not reporting a belief about the answer; it is completing the probe suffix in a plausible format while its reasoning continues underneath. The correct value $\frac{1}{8}$ appears only at the final probe. Agreement here is maximal and perfectly uninformative, which is why any deployed rule needs to check answer *shape* before treating agreement as evidence (§5.3).

Case 3: when there is nothing to recover. The minority case is equally worth stating. Problem 240 (gold 116) gives

$$52 \times 2, 104, 52 \times 5, \dots, 154 \times 8, 52, 154 \times 62$$

A three-probe rule fires at token 192 and commits 52; the trajectory itself ends at 154. Both are wrong. Here early stopping destroys nothing, because continued reasoning was not going to arrive at the right answer either—the model settles firmly on 154 and holds it for its final 62 probes. Cases like this are why the harm-to-rescue ratio of §5.4

Selection → generalization: consensus train candidates (orange) leave the gate on dev/test; the three DEER operating points (C/B/T) stay in it

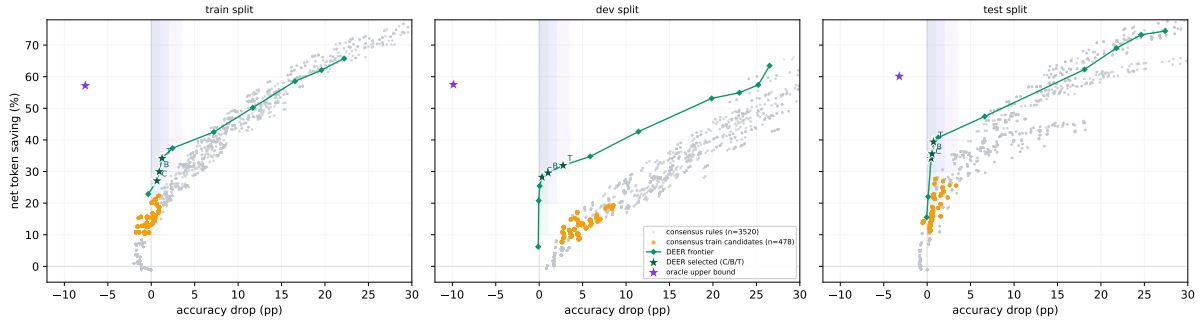


Figure 7: Selection → generalization across splits (two development models; macro-averaged over environments). Grey: all 3,520 consensus rules. **Orange**: the 478 rules that clear the conservative gate *in-sample on train*. **Green**: the DEER threshold frontier, with the three dev-selected operating points (Conservative/Balanced/Token-efficient) as stars. **Purple star**: the oracle upper bound (earliest correct probe).

counts only stops where committing *changes* correctness: on problems the model was never going to solve, a consensus stop costs accuracy nothing and saves tokens, which is the regime consensus early exit was designed for. The difficulty is that agreement does not distinguish this case from Case 1, and Case 1 is the more common one.

G False Consensus: Full Analysis

Section 4.1 summarizes, in four facts, what this appendix establishes in full: in an intervention-free setting, intermediate probe consensus is not a reliable stopping signal.

G.1 Setup

We analyze the frozen main trajectories (Section 3): DeepSeek-R1-Distill-Qwen-7B on all 500 MATH500 problems (Lightman et al., 2024; Hendrycks et al., 2021) at the main 16K-token budget, temperature 0.6, top- p 0.95, across three seeds (1,500 trajectories in total).⁴ The dense probe bank appends the boxed-answer probe suffix every 64 generated tokens and reads off the model’s current answer. The probe is a separate forward pass: the controller only *observes*, never halting or altering the main trajectory. This yields, per trajectory, the full answer sequence a stopping rule would have access to at the budget used throughout the paper.

⁴Train/dev problem ids are collected at seeds 42/43/44 and the held-out test ids at seeds 45/46/47; their union covers all 500 problems. This is a descriptive characterization of the frozen trajectories—it selects and tunes nothing—so it does not touch the preregistered sweep or the test-split commitment.

G.2 Final agreement is reliable; early agreement is not

Agreement measured at the *end* of a trajectory does track correctness. On the 186 trajectories that reach unanimous *cumulative* agreement (all non-empty probes concur), 97.8% are correct—only four whole-trajectory false consensus; and of the 1,205 whose *final five-probe window* is unanimous, 90.4% are correct. But this end-state agreement is not what an online stopping rule can act on: a rule fires the *first* time it sees agreement, mid-trajectory, long before the answer has settled. The rest of this appendix shows that *early* agreement—the signal a controller must actually use—is far from terminal.

G.3 The cost lands at the stopping decision

We simulate a *naive consecutive-agreement* stop: halt at the first point where three consecutive probes agree, are non-empty, and are *certain* (the model emits the boxed answer with high token-level confidence; formal definition in Appendix A). This is a simplified heuristic, *not* the CertIndex (CoT) rule, which we reproduce from its released implementation in §4.4. It fires on 1,477/1,500 trajectories. The committed answers are correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%—a 40.2 pp accuracy loss in exchange for an average saving of 3301 tokens, with 731 trajectories stopped on an outright wrong answer.

G.4 Recovery is common, and early stopping kills it

The loss is driven by *recovery*: continued reasoning overturns the intermediate answer far more often

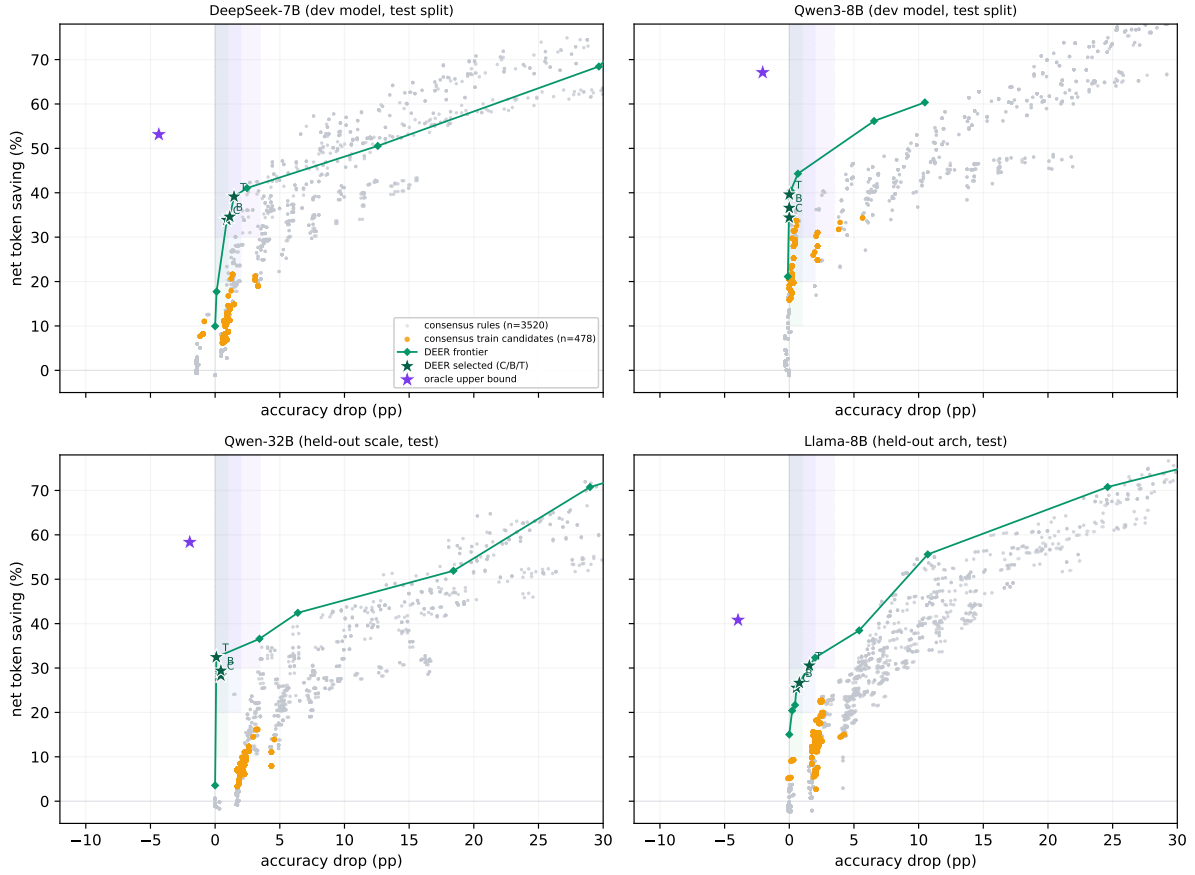


Figure 8: Out-of-sample generalization across scale and architecture (all panels test split). The two development models (top) are shown on their held-out test seeds; the two held-out models (bottom) are entirely unseen. Markers as in Figure 7. The dev-selected DEER points stay inside the conservative gate on every model; no consensus rule does.

than intuition suggests. Of the 736 trajectories that at some point formed a three-probe consensus *different* from their final answer, 620 (84.2%) ended up correct; and of the 1,148 whose *first* probe was wrong, 89.1% eventually flipped to correct. Neither “first impression” nor “local consensus” is close to terminal.

Counterintuitively, *later* consensus is *less* trustworthy: consensus formed before 512 tokens is 91.0% correct, but consensus that only forms after 2048 tokens is 71.6% correct. Hard problems both converge slowly and converge to worse answers—exactly the problems on which an aggressive stopping rule does most damage.

G.5 An error taxonomy

Hand-inspecting stopped-but-wrong cases (a preliminary AI-assisted labeling of 28 cases from an earlier exploratory pass, not the automated counts

above) yields *four* non-empty types out of five categories. By count: **(A)** numeric collapse—stable convergence to a wrong number—14/28; **(D)** derivation gaps—dropped roots/cases, unverified solutions—7/28; **(E)** format/option hallucination—a non-multiple-choice problem stably emitting a letter such as “B”—6/28, which is an artifact of the probe mechanism itself rather than a model belief; and **(B)** expression collapse—1/28. Type **(C)** (sign errors) did not occur in this sample. Type E is itself a methodological finding: roughly one in five “false consensus” is a probe formatting artifact, implying that any deployed rule should filter answer *shape*.

Takeaway. These observations—local consensus unreliable, recovery common, late consensus worse—suggest that a good stopping rule must be much more than “agreement.” The rest of the paper

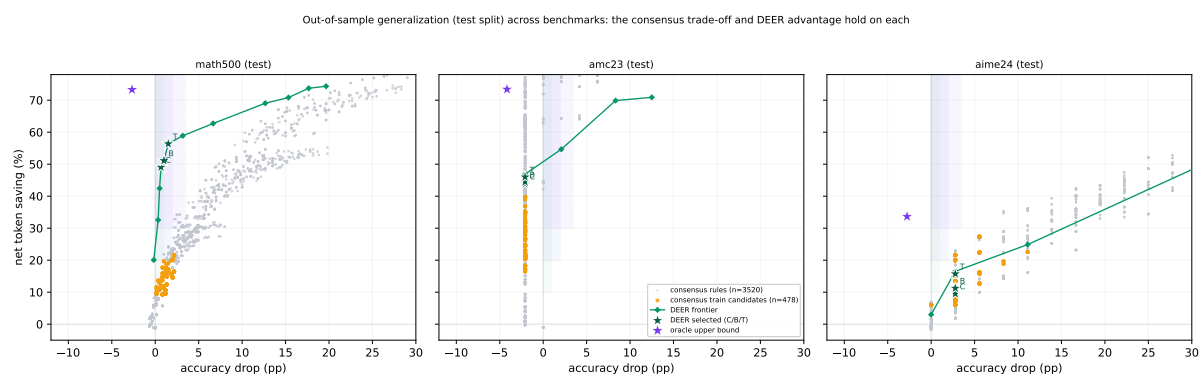


Figure 9: Out-of-sample generalization across benchmarks (test split, development models). The consensus trade-off and the DEER advantage hold on MATH500 and AMC23; on the small AIME24 set (few problems per environment) the two frontiers are noisier and intertwine. Markers as in Figure 7.

asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical. In the searched space, it cannot.